Understanding the Relationship Between Promoter Methylation and TP53 Gene Expression Using UCSC Genome Browser.

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Manual

Purpose Statement:

This project asks: **Does DNA methylation at the TP53 promoter vary across human tissues in ENCODE datasets and does promoter methylation correlate with TP53 expression?**
Using the UCSC Genome Browser, I will demonstrate the approach for visually integrating methylation (WGBS) and RNA-seq tracks to generate hypotheses about epigenetic regulation of TP53. This gene was focused on due to its importance in identifying damaged DNA and protection cells against cancer generation.

UCSC Genome Browser: https://genome.ucsc.edu/index.html

Hyperlink to Video Tutorial: https://youtu.be/CTWAAaLzKrI

Hyperlink to Presentation: https://youtu.be/H-Z96YbYuOg

Background

What is UCSC?

The UCSC Genome Browser is a publicly available tool that is widely utilized for its interactive database. This database allows the user to visualize a wide range of genomic annotations like methylation data and expression tracks. The platform gives precise molecular characteristics and features for specific genomic regions of interest. UCSC also has the ability to overlay data to perform comparative analyses.

Why do we use it and how is it different?

UCSC helps to observe certain patterns like DNA methylation and RNA-seq tracks for multiple tissues. The genome browser can pull readily available information from other publicly available databases like the ENCODE Project to visualize assays like Whole-Genome Shotgun Bisulfite Sequencing (WGBS) and RNA-seq. The ENCODE Project will be utilized due to its wide range in information available regarding the human genome. From this data, UCSC allows the user to identify the correlation between different gene features to perform further quantitative analyses by exporting data elsewhere.

The widely diverse set of high-quality annotation tracks make the UCSC Genome Browser stand out from others. In alternative browsers, like Ensembl, the main feature is to view certain aspects like gene ID and transcript ID, which don't allow you to visualize the actual genomic information along related variants. UCSC allows the user to customize their datasets with different types of visualizations and filters. It also contains a large scale of references for comparative genomics like CpG islands, predicted genes, homologies, mRNA, and so many more (Kent et al., 2002). However, UCSC can potentially be overwhelming for beginning users, as it's visualizations can become easily confusing. This can make the display cluttered and more challenging to interpret data. This document contains a step-by-step manual to demonstrate the use of UCSC to determine DNA methylation and expression characteristics of the TP53 gene.

How will it help answer the big-picture question?

The big question is: **Does DNA methylation at the TP53 promoter vary across human tissues in ENCODE datasets, and does promoter methylation influence TP53 expression?** To address this, I will utilize UCSC to visualize publicly available data from the ENCODE Project directly in the genome browser. ENCODE uses a vast range of techniques like RNA sequencing, DNA methylation assays, comparative genomics, and chromatin immunoprecipitation assays. In this manual, methylation tracks taken from Whole-Genome Bisulfite Sequencing (WGBS) and RNA-seq expression tracks from the same cell types (GM12878, lymphoblastoid and HeLa-S3, cervical cell lines) will be displayed (taken directly from ENCODE) alongside correlating genomic features including CpG islands, transcription start sites (TSS), and promoter regions (Rosenbloom et al., 2012). CpG islands are regions with a high frequency of CpG dinucleotides.

These are usually unmethylated and are observed near the promoter region in around 70% of vertebrate genes. Hypomethylation of CpG islands are typically associated with active gene expression, whereas hypermethylation can lead to gene silencing (Illingworth and Bird, 2009). Aligning these datasets within UCSC will enable a direct visual assessment as to how promoter methylation corresponds to the expression intensity across the chosen tissues. The co-visualization of these datasets can reveal potential relationships like a higher promoter methylation correlating to a lower TP53 RNA-seq signal.

Methylation Patterns and the TP53 Gene

DNA methylation patterns, specifically near the promoter region of a gene, are important in understanding gene expression because higher levels of DNA methylation can "turn off" a gene. Methyl groups, specifically 5-methylcytosine (5mC), can reduce transcription factor binding and expression by physically block a gene, particularly its promoter region, so it cannot be activated for its specific task. The TP53 gene encodes the p53 protein, which is a crucial tumor suppressor gene that helps to regulate cell cycle arrest, DNA repair, and apoptosis in response to cellular stress. Abnormal hypermethylation of the TP53 promoter has been shown to reduce its transcriptional activity, meaning that it becomes silenced and can be a huge contributor to tumor growth (Ng and Yu, 2015). Understanding the relationship between TP53 promoter methylation and its expression across different cell types in UCSC can provide valuable insight towards the impact of epigenetic regulation on cancer-related gene function.

Tools/Datasets

To address the proposed research question, this project uses the UCSC Genome Browser as the primary analysis database. UCSC will help to integrate genomic tracks, along with datasets from the ENCODE Project to help understand variables like DNA methylation, RNA expression, and transcription factor binding across different human tissues.

UCSC Genome Browser: https://genome.ucsc.edu/index.html

- Interactive, web-based genome visualization platform
- Integrates data from projects like ENCODE, Roadmap Epigenomics, or GTEx
- Reference genome assembly: GRCh38/hg38 (Genome Reference Consortium Human Build 38)
  - UCSC internal name: hg38
  - Assembly accession: GCA_000001405.29

These following datasets are publicly available as hosted tracks within UCSC's ENCODE track hub. These datasets can be directly visualized through the "Mapping and Sequencing", "Regulation", and "Expression" tracks.

Assay Type	Cell Line	ENCODE ID	Biological Source	Description
WGBS	GM12878	ENCSR890UQO	Human B lymphoblastoid	Methylation percentages at CpG sites (5mC)
WGBS	HeLa-S3	ENCSR550RTN	Human cervical carcinoma cell line	Methylation percentages at CpG sites (5mC)
Poly A RNA-seq	GM12878	ENCSR000COQ	Human cervical carcinoma cell line	Expression data for transcript abundance
Poly A RNA-seq	HeLa-S3	ENCSR000CPR	Human cervical carcinoma cell line	Expression data for transcript abundance

Table 1. Datasets for UCSC Analysis

Note: The GM12878 and HeLa-S3 cell lines were chosen due to their normal, or "wild-type" (WT), function in the TP53 gene. Abnormalities or mutations in either of these cell lines can directly influence the activity of TP53 and have been widely utilized in cancer studies.

Manual

This manual serves as a guide for navigating the UCSC Genome Browser to visualize DNA methylation patterns and gene expression at the TP53 locus. It will demonstrate how to locate the region of interest, enable specific ENCODE tracks for co-visualization, and interpreting data for downstream analysis. This will help gain more knowledge regarding the correlation between methylation patterns and TP53 expression across the selected tissues.

Step 1: Accessing the UCSC Genome Browser

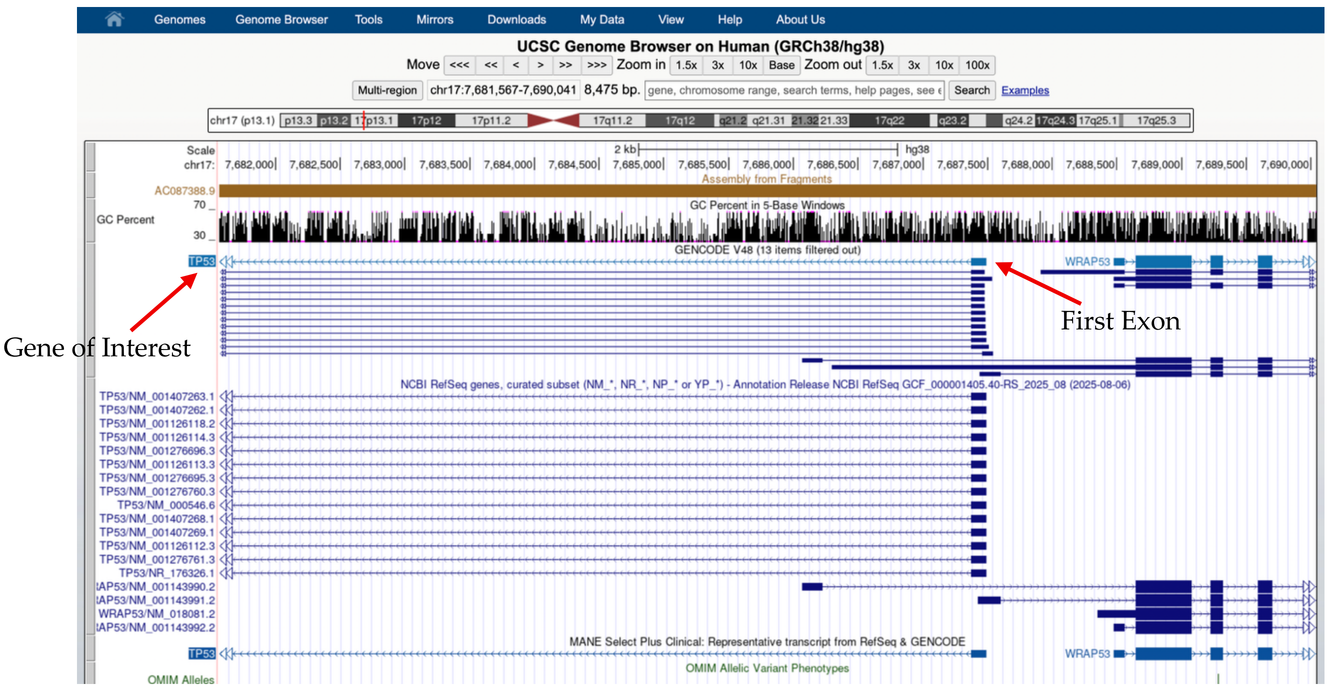
1. Open any web browser (Chrome, Safari, etc.) and go to https://genome.ucsc.edu.
2. On the homepage of the database, select "Genome Browser" under the *Tools* menu (it should be in blue writing).
3. In the *UCSC Species Tree and Connected Assembly Hubs*, make sure "Human" is selected.
4. Apply the following settings before clicking search:
    a. Under *Human Assembly*, select "Dec. 2013 (GRCh38/hg38) assembly.
5. Click *Go* to launch the main browser view.

Here, you are observing the human reference genome assembly chosen. The horizontal tracks in the middle portion of your screen represents the different gene models and various annotations layers.

Step 2: Locating the TP53 Gene

1. In the search bar at the top (located next to the chromosome coordinates), type in "TP53" and press *Go*.

2. From there, UCSC will redirect you to the TP53 genomic locus. At first glance, you can notice that this gene is located on chromosome 17 as displayed in the upper panel of the browser.
    a. To ensure the exact position in which this gene is located, you can look at the coordinates towards the top of your screen. It should look like this:
        i. "chr17:7,668,421-7,687,490"
3. Begin zooming in and out using the navigation bar until the promoter region and first exon are clearly visible.
    a. This will be located towards the right end of the gene, since it is located on the minus strand. The presence of an exon is indicated by a thick blue box, also indicated by the red arrow shown below.



You should now be able to examine genomic features like CpG islands and promoter methylation in relation to transcription start site (TSS).

Step 3: Enabling CpG Island and Annotation Tracks

1. Scroll down past the genomic models and find the different track sections. Click on the *Regulation* track.
2. Locate "CpG Islands" and set it to "show".
3. Refresh that track to visualize the CpG-rich regions. Those regions will be displayed as green bars across the gene of interest.
4. Under the *Genes and Gene Predictions* track, set the "NCBI RefSeq" filter to "dense" and refresh.

    a. This will allow you to visualize the exons (thick blue boxes), introns (thin blue lines), and transcript start/end sites (TSS/TTS).
5. Also under the same track, set the GENCODE V48 filter to "full" and refresh.
    a. This will show any alternative, Ensembl-based transcripts.
6. *Disclaimer: Both RefSeq and GENCODE capture similar data, but RefSeq is better for human genetics/variant annotation, whereas GENCODE is more useful for comprehensive analysis, as it captures a wider range of transcripts.

Note: Higher CpG island activity is typically observed as clusters near the promoter region of the TP53 gene. The methylation patterns within these areas can help provide insight towards the transcriptional regulation of this gene.

Step 4: Adding ENCODE DNA Methylation Tracks

1. In the menu at the very top of the screen, click "My Data → Track Hubs".
2. Under Public Hubs, locate "ENCODE DNA Trackhub", and click connect.
    a. This will load a new hub in your list of tracks, and it should include a filter called "whole-genome shotgun bisulfite sequencing".
3. Click on this filter and locate the two data types of interest. These are the specific cell lines we are looking at:
    a. The lymphoblastoid: GM12878
    b. The cervical carcinoma cell line: HeLaS3
4. Click the + button next to each of these cell lines (only for the ones that say "CpG"), scroll down, and ensure that each of the selected cell lines are those two, and they each are set to "full".
    a. You should have a total of four lines selected, each one representing the plus and minus strand of that cell line.
5. Click submit, and this will bring you back to your genome browser. You might get a warning error saying that your image was too large, and that it will display them as density graphs. To overcome this, you can simply zoom in to toggle your view on the individual cell lines.
    a. To avoid this, you could set them to "dense", and it will compact how they are visualized.

6. You’ll notice above that when you hover your mouse over an individual methylation track, it will read a specific number value, shown in the image above. This value indicates the percent methylation of the cell line at that specific position. The example shown above reads 0.637553 at a region where a CpG island is present (you can confirm this by scrolling down to make sure there is a CpG island lined up with your mouse).
   a. This value tells you that at that precise location, ~ 63.8% of the total reads are methylated.
7. Drag your mouse across each of these cell lines. You’ll notice that there are higher peaks and higher percent methylation values in regions where CpG islands are present.

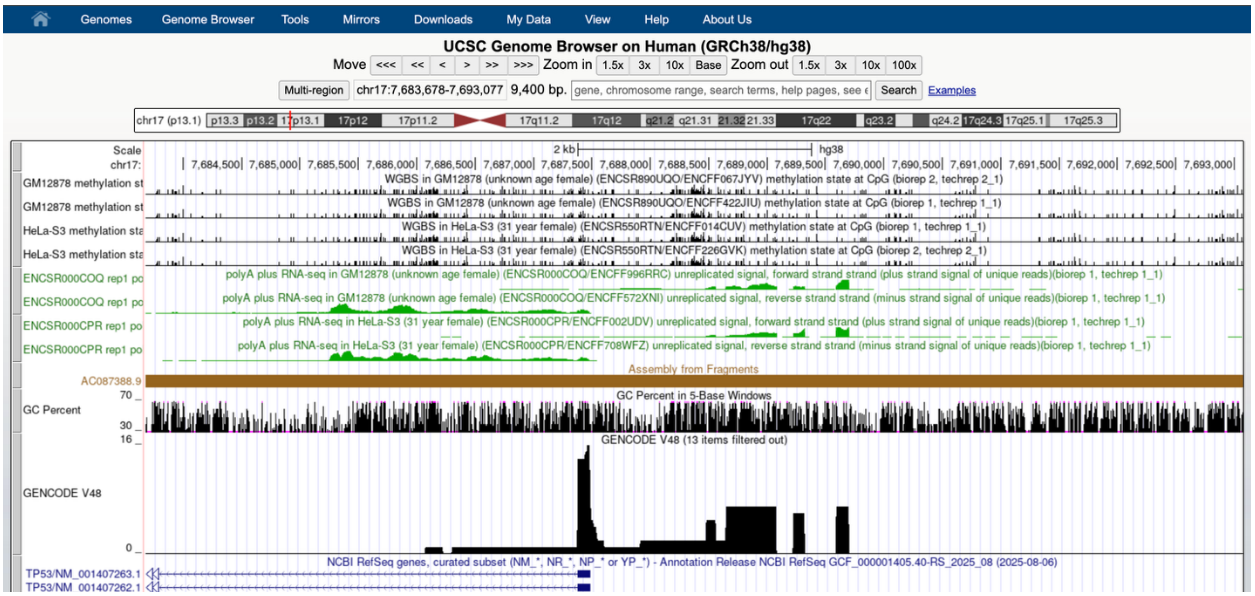
By visually comparing these two cell types, you can detect regions on TP53 that are more heavily methylated than others.

Step 5: Adding RNA-seq Expression Tracks

1. In the menu at the very top of the screen, click “My Data → Track Hubs”.
2. Under the “Search terms” tab, type in “RNA-seq”, and ensure that your assembly is the hg38 one, and click “Search Public Hubs”.
3. In the list of available hubs, you should find one labeled “ENCODE RNA Trackhub” and click Connect.
   a. You should now see it added as a new track in your browser.
4. Locate the “RNA-seq (by biosample)” filter and click on it.
5. Click on where it says “RNA-seq (by biosample) GM12878”. It will redirect you to a list of sub tracks you can display in the browser. Under “minus strand signal replicate 1”,

find the ENCODE ID *ENCSR000COQ*. Remove every other subtrack by clicking on the pink checkmark on the left.

6. Do this same thing for the "plus strand signal replicate 2", ensuring only that one ENCODE ID is selected.
    a. Click Submit.
7. Repeat steps 5-7, but making sure you click on "RNA-Seq (by biosample) HeLa-S3".
8. After clicking Submit, you should be able to see both cell lines, along with the Poly A RNA-Seq analysis data.
9. It should look something this this in your model:



The RNA-seq tracks display transcriptional activity as coverage plots. Similar to methylation patterns, RNA expression is represented by the little blue boxes that pop up when you hover over that specific position. Higher values and higher peaks are indicative of more RNA transcripts mapped to that region. Comparing the expression tracks with methylation patterns can reveal a potential relationship between promoter methylation and gene activity.

Step 6: Focusing on the Promoter Region

1. Zoom to the region surrounding the TP53 transcription start site (TSS).
    a. This will be around 7,687,500-7,688,000 bases.
2. Examine how CpG islands overlap with methylation peaks.
3. Compare RNA-seq signal intensity in the two cell lines (refer to Step 5.8).

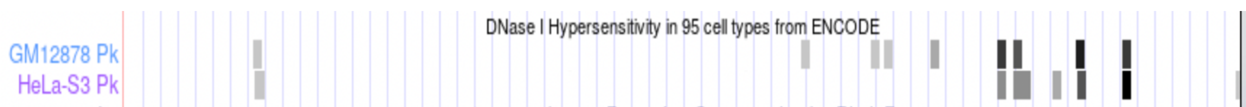
If methylation suppresses transcription, the HeLa-S3 track might show higher methylation near the promoter and lower RNA-seq signal, whereas GM12878 may display lower methylation and higher RNA expression. This pattern reflects how promoter methylation can act as an epigenetic mechanism to silence gene expression in specific tissues/cell types.

Step 7: Adding Functional Annotation Tracks (optional but helpful)

Adding DNase I Hypersensitivity:

- This is a filter provided by ENCODE which marks open chromatin regions accessible to transcription factors. This can be a useful indicator of where regulatory proteins might bind to the gene.

1. Locate the *ENCODE Regulation filter* under the "Regulation" track.
2. Click on it, and then click on "DNase HS".
    a. Under Cell Type, select our two cell lines of interest, GM12878 and HeLa-S3.
3. Ensure they are set to "full" and click Submit.
4. You should then be able to the view regions (black and grey boxes) that are more hypersensitive, indicating that DNA is not wrapped tightly around nucleosomes. The more hypersensitive a region is, the more accessible it is to transcription factors.
5. The cluster of black boxes observed across both cell lines indicates a regulatory hotspot for transcription.
6. The grey boxes indicate the signal intensity, so how strong the DNase-seq read coverage is.



Adding H3K4me3 Histone Modification:

- This is a filter which is also provided by ENCODE that can demonstrate histone modification present near active promoters. Locate the *ENCODE Regulation filter* under the "Regulation" track.

1. Locate the *ENCODE Regulation filter* under the "Regulation" track.
2. Click on it, and then click on "Layered H3K4Me3".
    a. You'll notice that there is no available data on the HeLa-S3 cell line.
    b. We will just focus on the GM12878 cell line, so make sure that is the only one selected, and click Submit.
3. You should now see peaks of H3K4Me3 marks, with the higher peaks indicating a potential strong presence of an active promoter. You'll notice that the peaks are relatively high throughout the entirety of the TSS of TP53.

Step 8: Exporting Quantitative Data

1. Locate and hover over the Tools tab at the top of your screen and select "Table Browser".
2. Ensure that you have the correct assembly – GRCh38/hg38 (this should already be selected since it's the one we've been looking at).
3. Under *Group*, select the "ENCODE DNA Trackhub", and the WGSB sequencing track.

4. since we now know where the promoter region is on TP53, we can set specific coordinates to focus on.
    a. Select "Position" and type in: chr17:7,661,000-7,688,000 (this does not need to be exact, so long as the range captures some information upstream and downstream of the promoter region).
5. Under *Output format*, select "BED – browser extensible data", and click Get output.
    a. This will display a table with several columns indicating various information regarding each position within the specified region.
6. Here is a key that defines each column:
    a. First column (e.g., chr17) – this is the chromosome where this feature is located.
    b. Second column (e.g., 7,661,039) – this is the start coordinate for that genomic position.
    c. Third column (e.g., 7,661,040) – the 1-based genomic end coordinate. This means that 1 base pair was covered at position 7,661,039.
    d. Fourth column (e.g., .) – this is a placeholder for the feature name (usually blank).
    e. Fifth column (e.g., 0) – this is the signal strength, representing the signal intensity. In this case, that is the methylation percentage. The higher the value, the stronger methylation was likely to have occurred at that position.
    f. Sixth column (e.g., + or -) – this indicates whether that region is located on the plus or minus strand.
    g. Seventh column (e.g., 7,661,039/7,661,040) – this column indicates the "thick" portion of the feature, meaning the region used in gene annotations or for understanding coding regions in other analyses).
    h. Eighth column (e.g., 0,255,0) – this column represents a specific color that encodes for different methylation levels.
        i. 0,255,0 = green – indicates very low signal, likely little to no methylation
        ii. 255,155,0 = orange – indicates slight signal, some methylation
        iii. 255,0,0 = red – indicates strong signal, likely high methylation
7. Do the same process for RNA-seq tracks to observe the expression values.
    a. For this, make sure to select the ENCODE RNA Trackhub.
    b. You will have to download the data separately for each cell line.

GM12878 Methylation

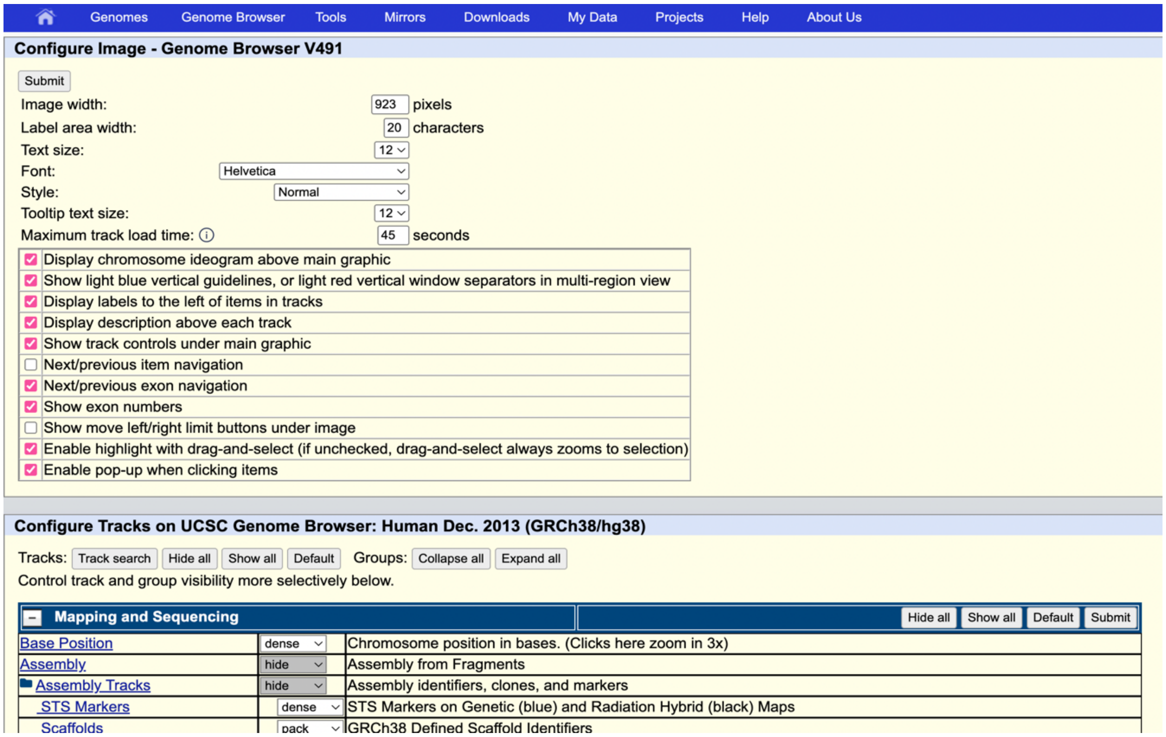
```
chr17   7661039 7661040 .   0   +   7661039 7661040 0,255,0
chr17   7661040 7661041 .   7   -   7661040 7661041 255,155,0
chr17   7661101 7661102 .   3   +   7661101 7661102 255,0,0
chr17   7661102 7661103 .   5   -   7661102 7661103 255,0,0
chr17   7661114 7661115 .   3   +   7661114 7661115 255,0,0
chr17   7661115 7661116 .   7   -   7661115 7661116 255,0,0
chr17   7661133 7661134 .   3   +   7661133 7661134 255,0,0
chr17   7661134 7661135 .   10  -   7661134 7661135 255,205,0
chr17   7661318 7661319 .   5   +   7661318 7661319 205,255,0
chr17   7661319 7661320 .   19  -   7661319 7661320 155,255,0
chr17   7661479 7661480 .   25  +   7661479 7661480 255,0,0
chr17   7661480 7661481 .   25  -   7661480 7661481 255,0,0
chr17   7661487 7661488 .   24  +   7661487 7661488 255,0,0
chr17   7661488 7661489 .   26  -   7661488 7661489 255,0,0
chr17   7661511 7661512 .   23  +   7661511 7661512 255,105,0
chr17   7661512 7661513 .   29  -   7661512 7661513 255,55,0
chr17   7661541 7661542 .   22  +   7661541 7661542 255,55,0
chr17   7661542 7661543 .   23  -   7661542 7661543 255,105,0
chr17   7661573 7661574 .   18  +   7661573 7661574 255,0,0
chr17   7661574 7661575 .   23  -   7661574 7661575 255,0,0
chr17   7661605 7661606 .   16  +   7661605 7661606 255,255,0
chr17   7661606 7661607 .   18  -   7661606 7661607 255,55,0
chr17   7661637 7661638 .   12  +   7661637 7661638 255,0,0
chr17   7661638 7661639 .   10  -   7661638 7661639 255,0,0
chr17   7661667 7661668 .   12  +   7661667 7661668 255,0,0
chr17   7661668 7661669 .   8   -   7661668 7661669 255,55,0
chr17   7661687 7661688 .   12  +   7661687 7661688 255,0,0
chr17   7661688 7661689 .   6   -   7661688 7661689 255,0,0
chr17   7661696 7661697 .   12  +   7661696 7661697 255,105,0
chr17   7661697 7661698 .   8   -   7661697 7661698 255,0,0
```

Step 9: Analyzing and Interpreting Results

1. Compare methylation levels – identify CpG sites with high methylation near the promoter.
2. Overlay expression data – observe whether these methylated regions are consistent with lower RNA-seq signaling.
3. Assess regulatory marks – high methylation coupled with the lack of expression from H3K4me3 might indicate some sort of promoter repression.
4. This allows us to make potential interpretations like:
    a. In HeLa-S3, promoter hypermethylation at the TP53 CpG island might reduce transcriptional activity of GM12878.

Step 10: Generating Figures (optional)

1. If you are using the UCSC Genome Browser for written analysis, you can create a PDF of your model to incorporate it into your results.
2. Hover your mouse over the *View* tab and select “PDF”.
3. To generate a PDF of just your model as it is, click on the “Download the current browser graphic in PDF”.
4. To create a more specific image for higher quality publications, you can add assembly names and chromosome range to your model. To do this, hover over the *View* tab and click “Configure Browser” and then select the desired configuration tracks as shown below.
    a. This will allow you to specify what gets added to your PDF. You can choose things like scale bar, assembly, zoom factor, and reverse complements of motifs.



Conclusion

This manual aims to illustrate the use of the UCSC Genome Browser as a powerful bioinformatics tool to **explore relationships between DNA methylation and gene expression at the TP53 locus**. Through the integration of methylation (WGBS) and expression (RNA-Seq), UCSC can allow for an efficient visualization of potential correlations between promoter methylation and transcriptional activity. Through the comparison of the two chosen cell lines, GM12878 (human lymphoblastoid) and HeLa-S3 (cervical carcinoma cells), users can observe how promoter methylation corresponds to TP53 expression changes. Qualitatively, the HeLa-S3 cell line displayed a higher methylation density across CpG-rich regions near the TP53 promoter. This can be supported by a lower RNA-Seq signal compared to GM12878, which might suggest a potential link between promoter methylation and transcriptional repression.

Though UCSC provides powerful visualizations, some limitations can occur that should be acknowledged. There is limited coverage of WGBS data in certain promoter regions, which shows the technical constraints that can arise due to the unavailability of certain datasets. This highlights the importance of integrating multiple evidence tracks to increase confidence in your data. Aside from those disadvantages, this project was able to demonstrate how UCSC serves as a useful and accessible platform for understanding genome-wide patterns. This bioinformatic tool is crucial as it can also offer insights towards epigenetic relationships of genes, like the tumor suppressor gene focused on in this demonstration.

Citations

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